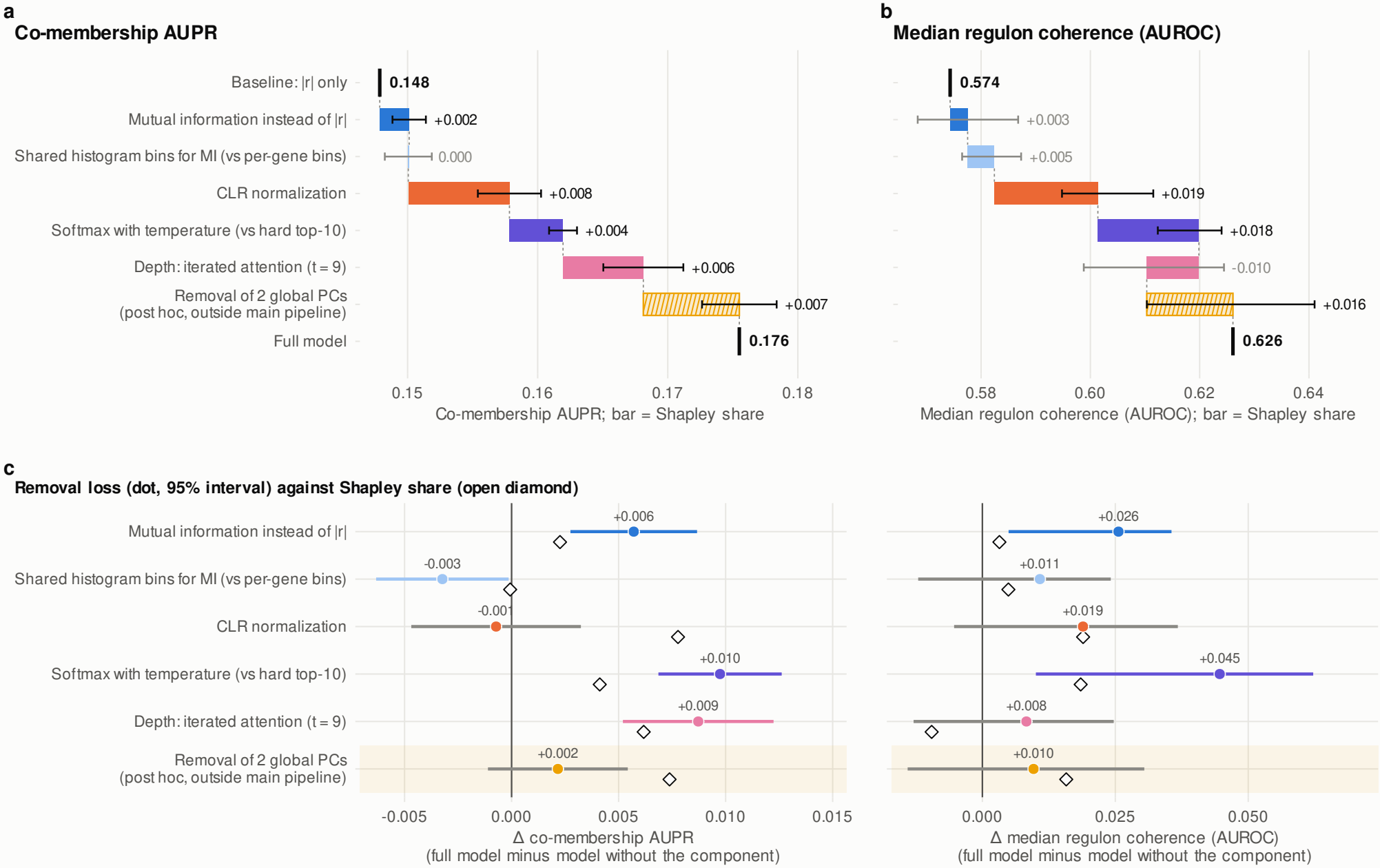
CLR normalization and softmax drive the coherence gain; CLR and depth drive pooled AUPR

M3D, 466 experiments; RegulonDB all evidence. a, b: exact Shapley shares over all 64 on/off combinations of six components, added in pipeline order from the |r| baseline to the full model; whiskers 95% intervals (gene jackknife for AUPR, regulon bootstrap for coherence), grey where they include 0. c: loss when one component is removed from the full model (dot, interval) against its Shapley share (open diamond); a gap is an interaction. Largest interactions: softmax on coherence (removal +0.045 vs Shapley +0.018) and mutual information on coherence (removal +0.026 vs Shapley +0.003): both matter most when the other components are present.



Hatched: PC removal was chosen after inspecting Supplementary Note 7 and is not part of the main pipeline. 'Off' alternatives: |r|, per-gene Freedman-Diaconis bins, raw dependence, hard top-10 neighbours, no attention, no PC removal. Where strong/confirmed (SC) and all-evidence results disagree in sign (interval excluding 0 at one level at least): Shapley AUPR for mutual information (SC -0.003, all +0.002); Shapley AUPR for shared bins (SC +0.003, all 0.000); drop-one AUPR for shared bins (SC +0.003, all -0.003); drop-one AUPR for CLR normalization (SC +0.006, all -0.001). All other estimates agree in sign or are within noise at both levels.